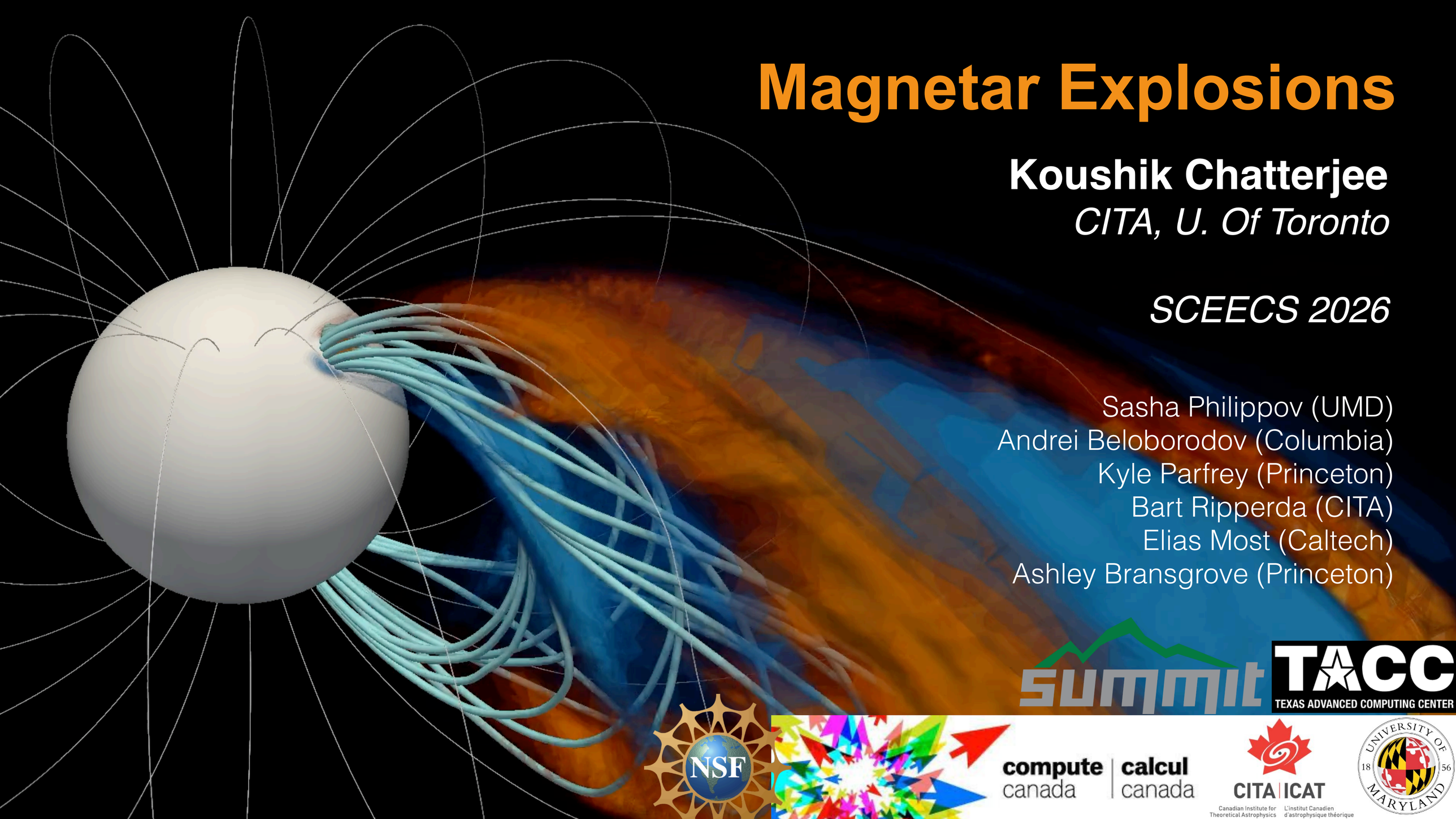


Magnetar Explosions

Koushik Chatterjee
CITA, U. Of Toronto

SCEECS 2026

Sasha Philippov (UMD)
Andrei Beloborodov (Columbia)
Kyle Parfrey (Princeton)
Bart Ripperda (CITA)
Elias Most (Caltech)
Ashley Bransgrove (Princeton)

summit

TACC
TEXAS ADVANCED COMPUTING CENTER

NSF

compute canada | calcul canada

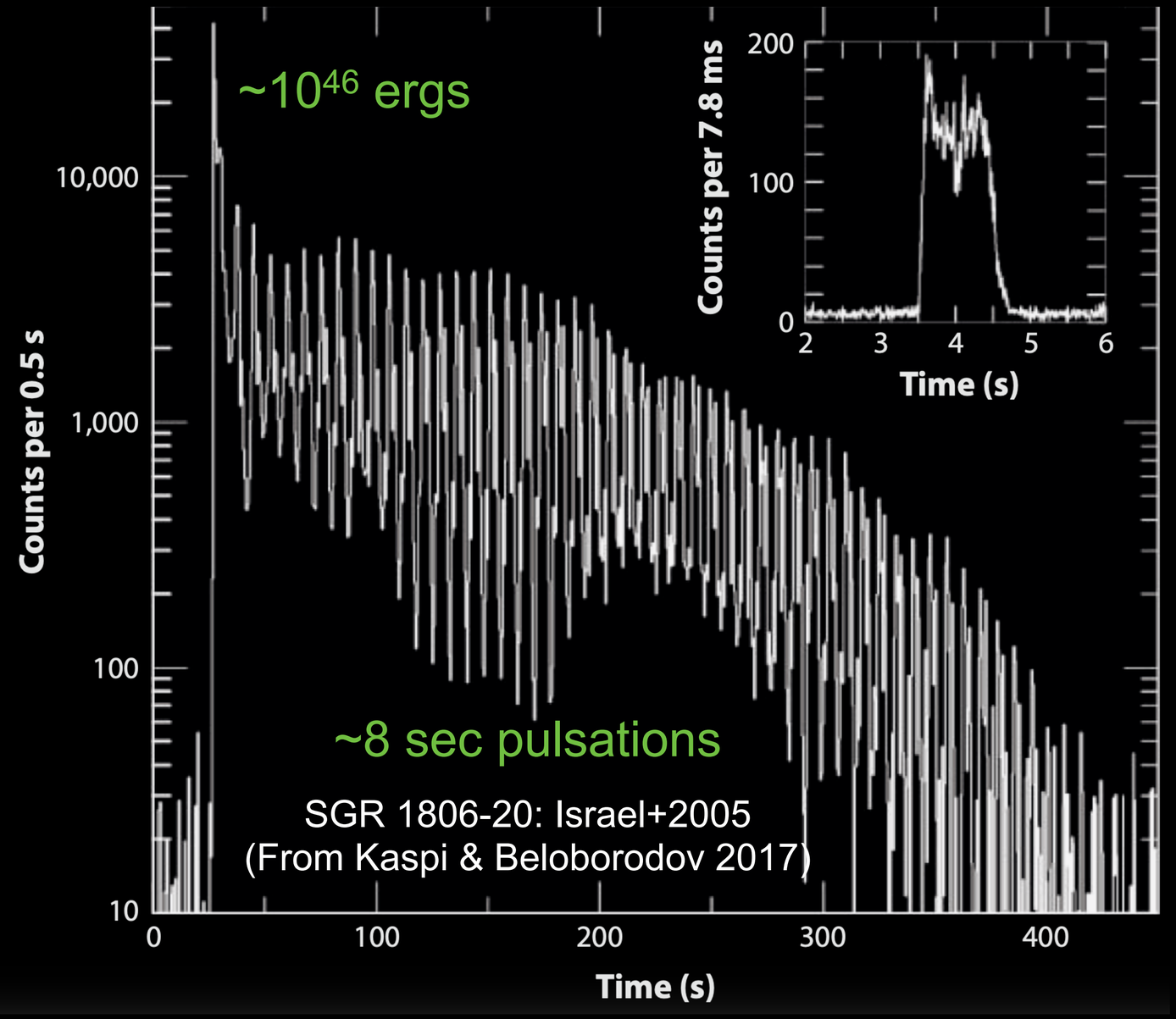
CITA | ICAT
Canadian Institute for Theoretical Astrophysics | L'institut Canadien d'astrophysique théorique

UNIVERSITY OF MARYLAND 18 56

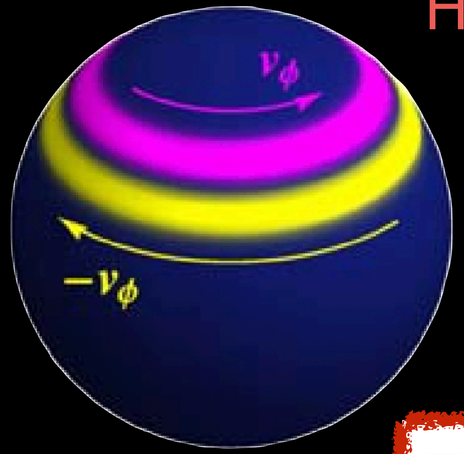
Magnetar Giant Flares

- Magnetars are slow-spinning (rotation periods ~ 10 sec), young NS with huge magnetic fields: B > 10^{13} G
- Giant flares are short, extremely bright X-ray emission. Only 3 detected: $10^{44} - 10^{46}$ ergs events
- Possible progenitors of Fast Radio Bursts (FRBs): cosmic sirens

Duncan & Thompson 92
Thompson & Duncan 95, 96
Lyutikov 06, ++
Thompson++
Beloborodov++



Magnetar Giant Flares: Crustal Twists



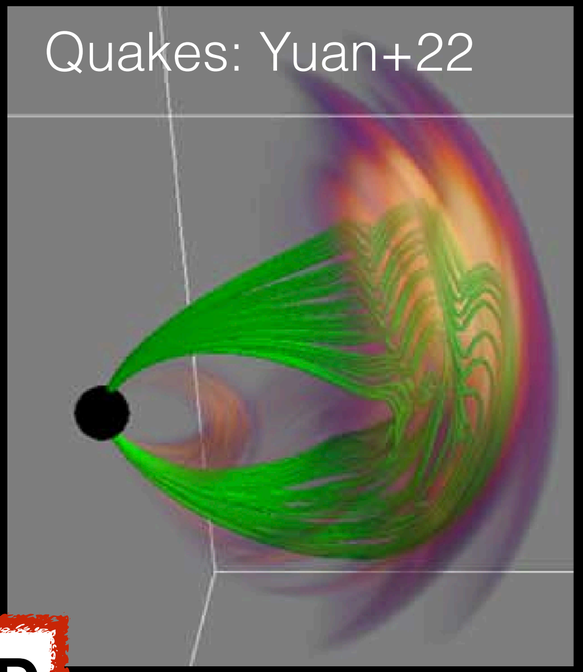
How do **Magnetar interiors** affect the NS environment?

Twist up surface magnetic fields and generate B_ϕ

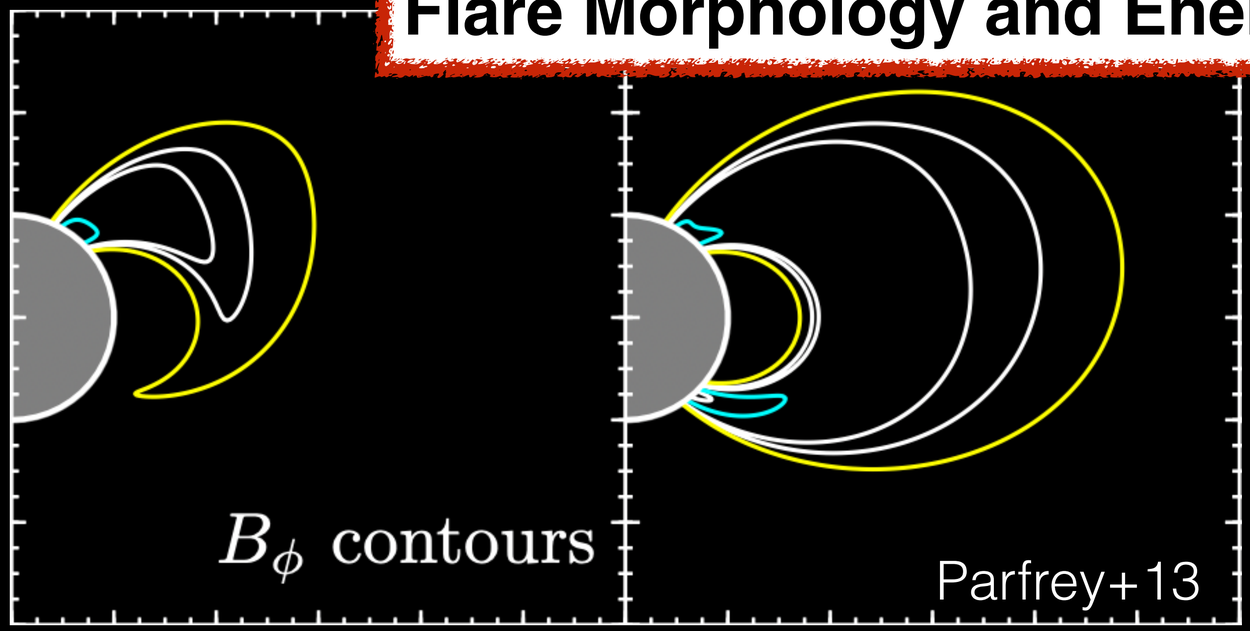
Inflate magnetosphere

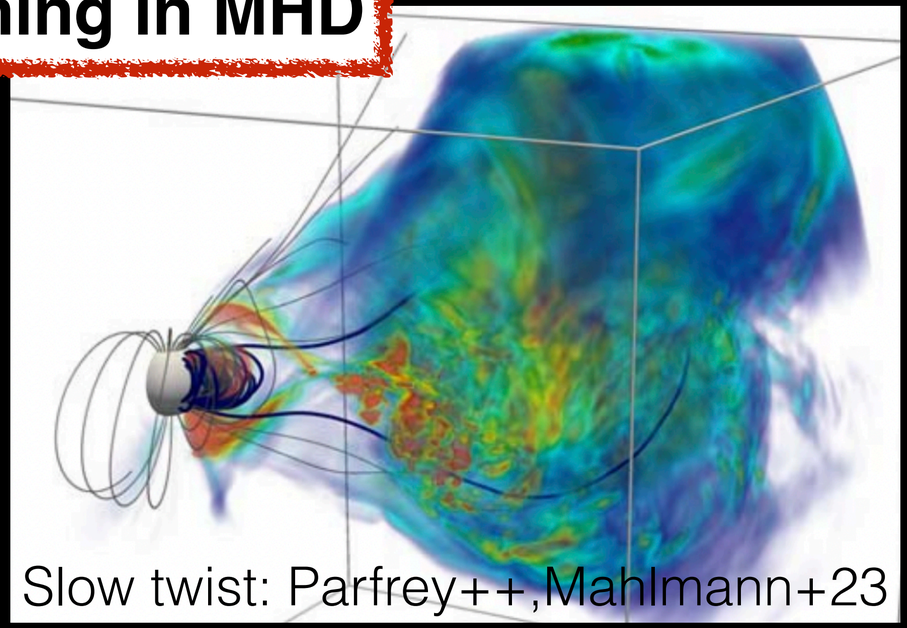
Unstable fluxtubes, reconnection

Ejected EM pulse ($\sim 10^{48}\,\mathrm{erg/s}$ for $10^{15}\,\mathrm{G}$ B-field)



Flare Morphology and Energy Partitioning in MHD



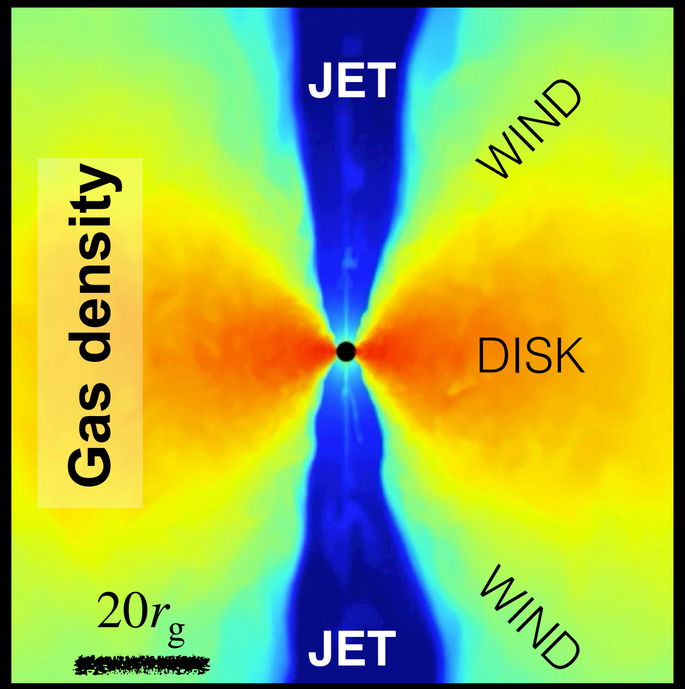


Theory and Numerics

General Relativistic Magnetohydrodynamics (GRMHD)

- Ideal collisional fluid: global accretion dynamics
- Does not capture non-thermal/kinetic effects

H-AMR (Liska, KC+22)



- GPU-accelerated
- Adaptive mesh refinement
- Local adaptive timestepping
- 2000^3 uniform grid accretion simulation on 6000 V100 GPUs *(Ripperda, Liska, KC+22)*

$$r_g \equiv \frac{GM}{c^2}$$

*Force-Free Electrodynamics (FFE)

- Highly magnetized plasma: jets, neutron stars
- Neglects plasma inertia and thermodynamics



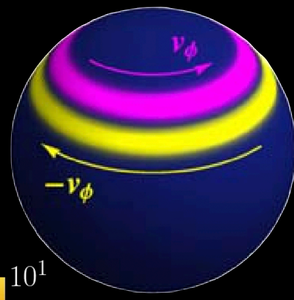
*FFE-GRMHD with *H-AMR (KC+26)*

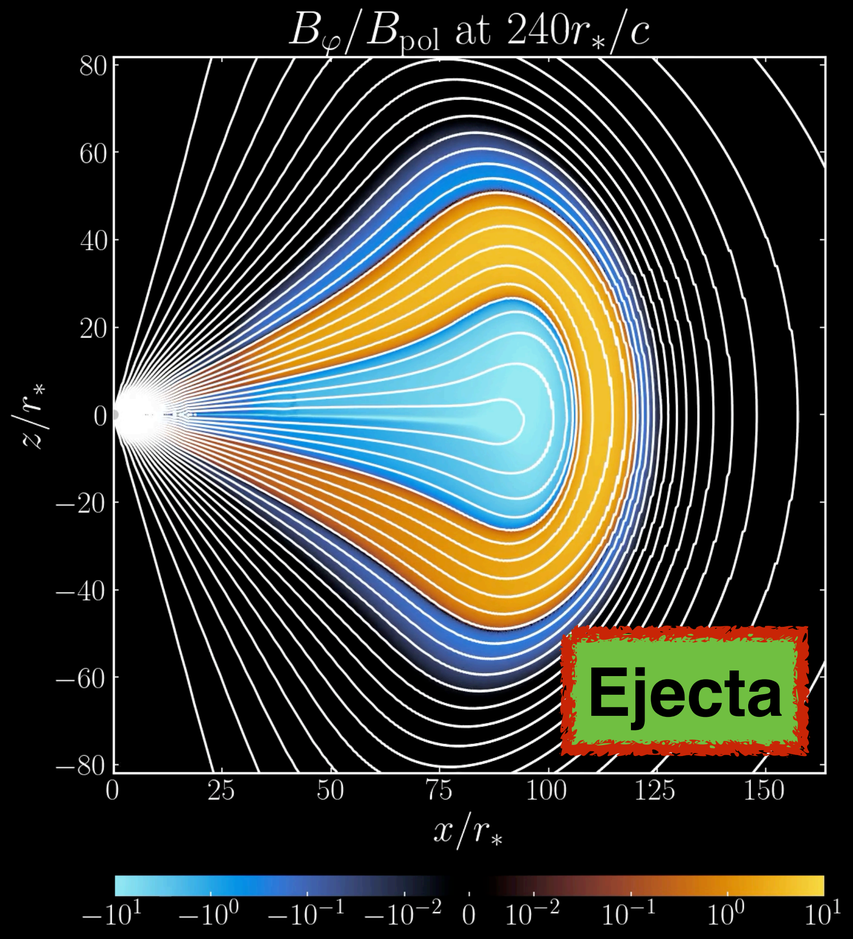
following methods from Parfrey & Tchekhovskoy 2017

Tackling Neutron Star Magnetospheres and Accretion (3D pulsars on 3000 GPUs)

SCEECS 2026

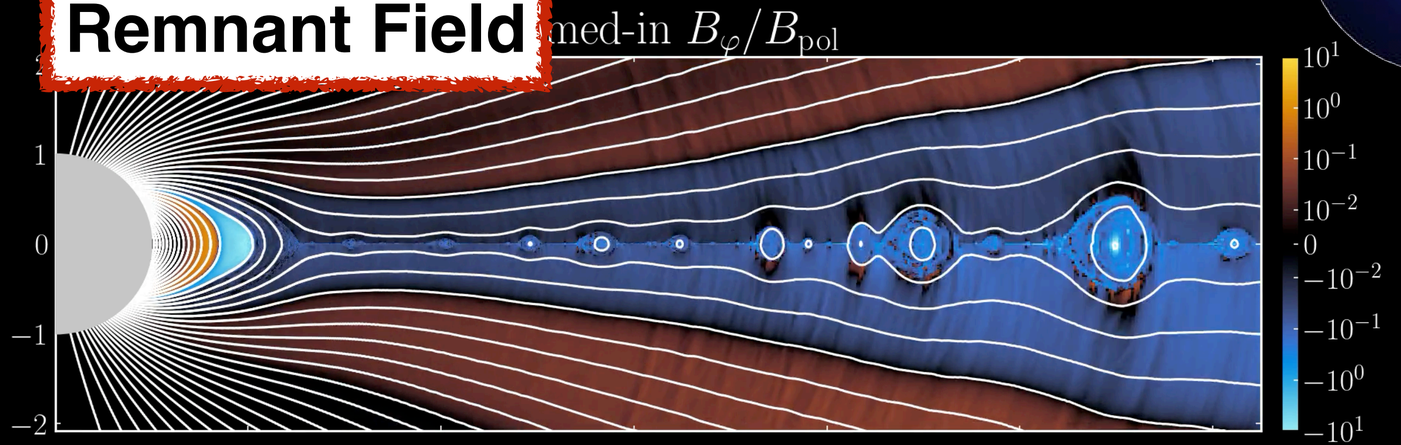
First MHD simulations of Magnetar Giant Flares

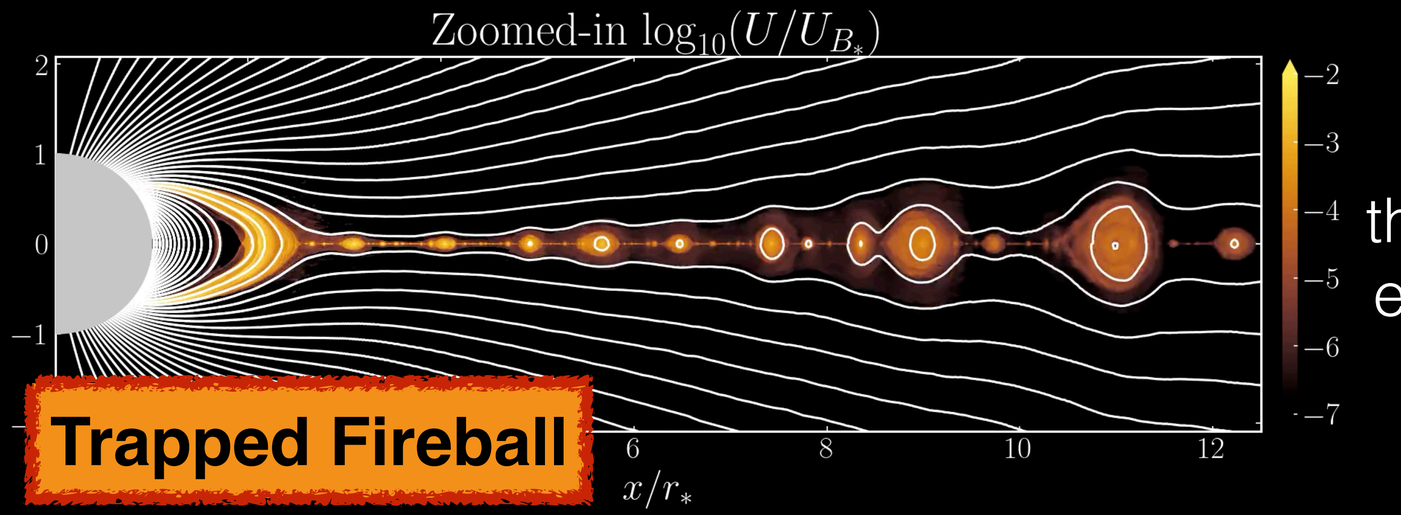




Ejecta

Remnant Field





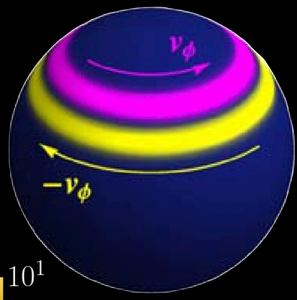
Trapped Fireball

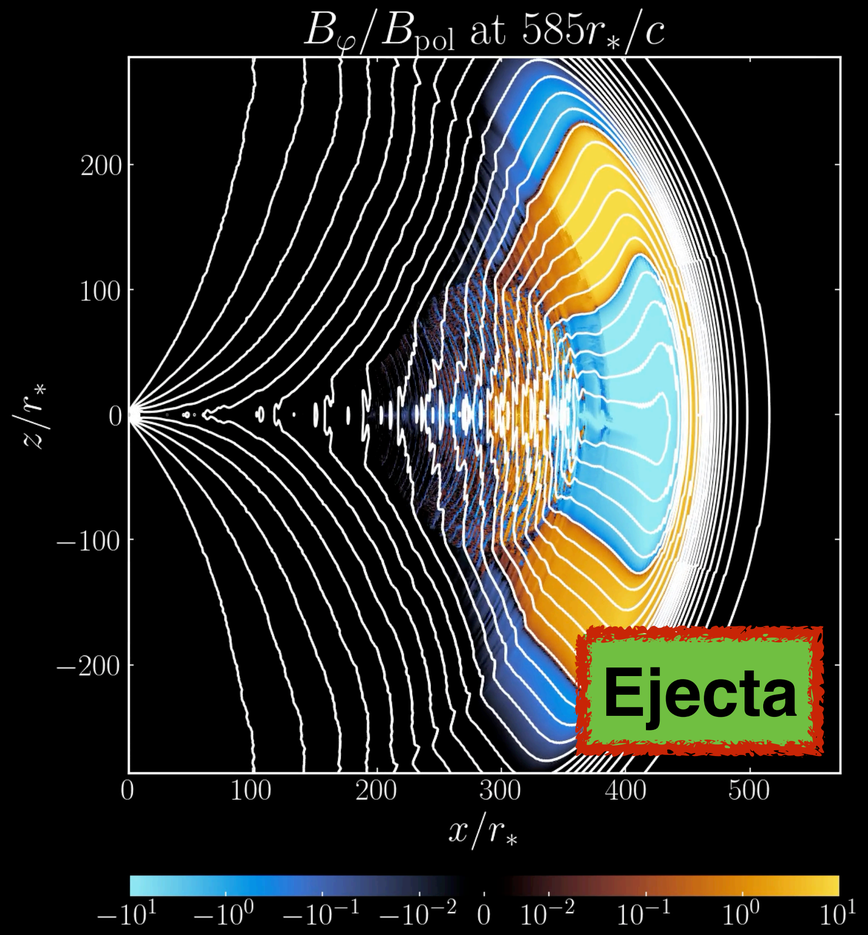
Fluid thermal energy

KC, Philippov, Beloborodov, Parfrey, Ripperda & Most 2026.

ICs: $\sigma_{\rm bkg} = 500, \beta_{\rm bkg} \lesssim 10^{-8}$

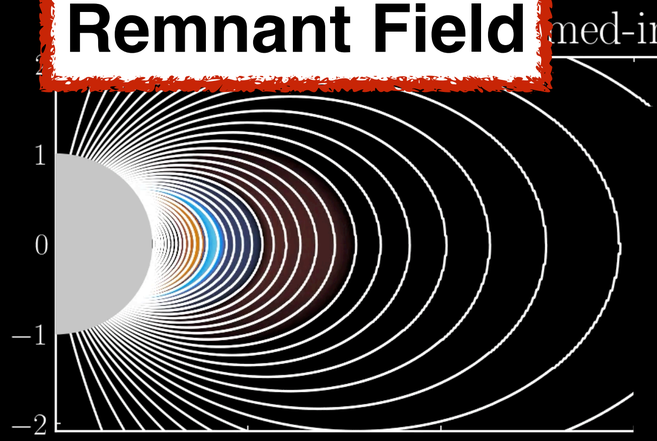
First MHD simulations of Magnetar Giant Flares





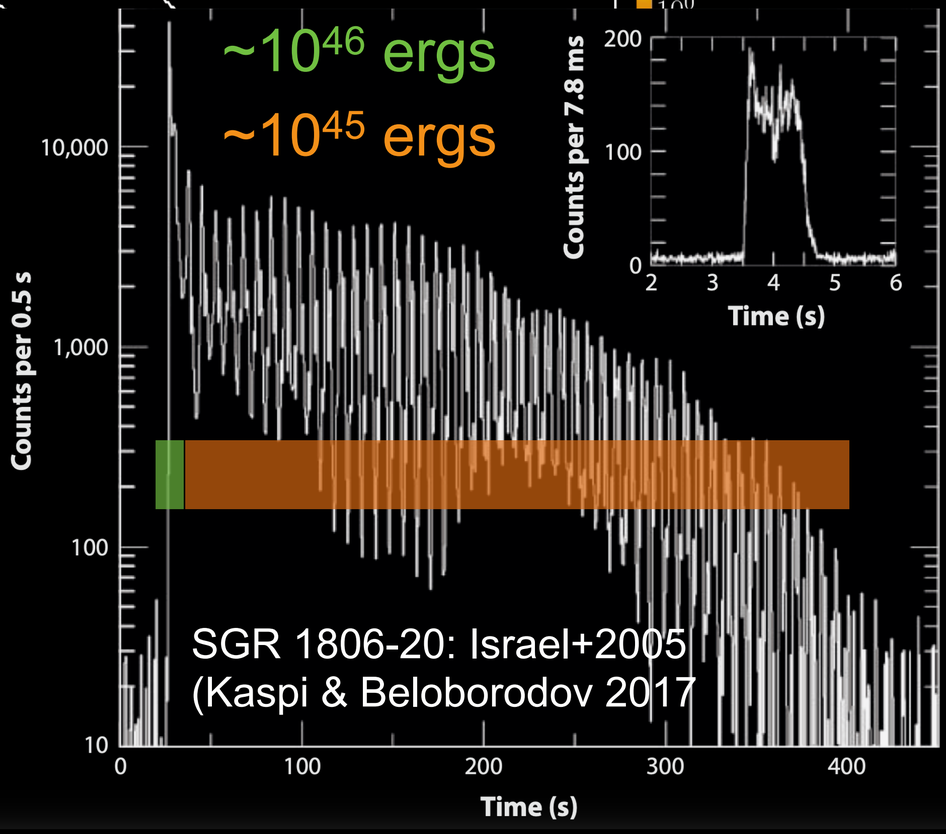
Ejecta

Remnant Field





Trapped Fireball



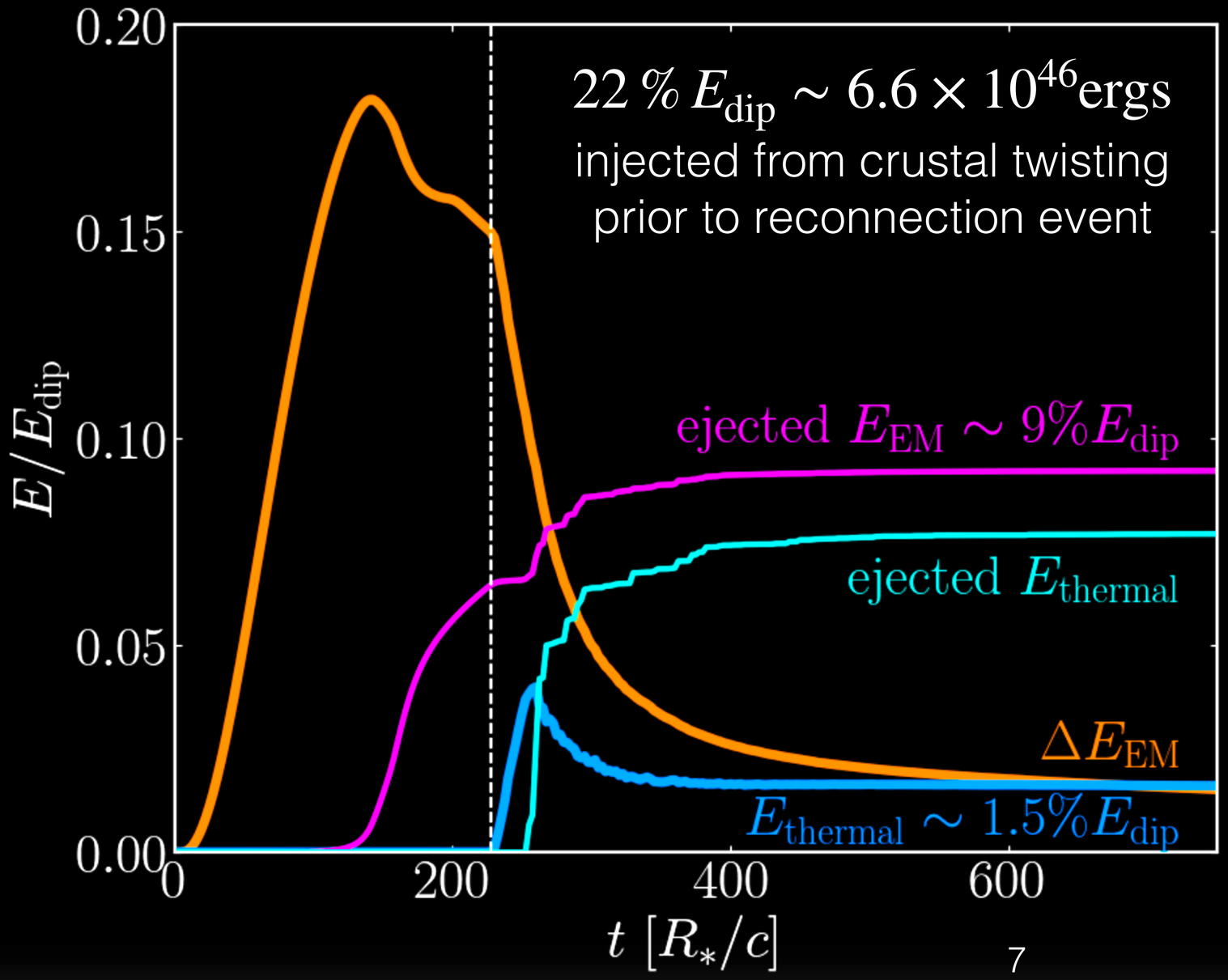
KC, Philippov, Beloborodov, Parfrey, Ripperda & Most 2026.

ICs: $\sigma_{\rm bkg} = 500, \beta_{\rm bkg} \lesssim 10^{-8}$

*assuming stellar B-field of 10^{15} G

Magnetar Giant Flares: Energetics

KC+26



$$E_{\rm dip} \sim 3\times10^{47}\left(\frac{B_*}{10^{15}\,\rm G}\right)^2\left(\frac{R_*}{10\,\rm km}\right)^3\ \rm erg$$

EM pulse ($\sim 10^{46}$ erg) can power FRBs ($\sim 10^{40}$ erg): *via shocks (Beloborodov+20), interactions with wind/current sheet (Lyubarsky+20)*

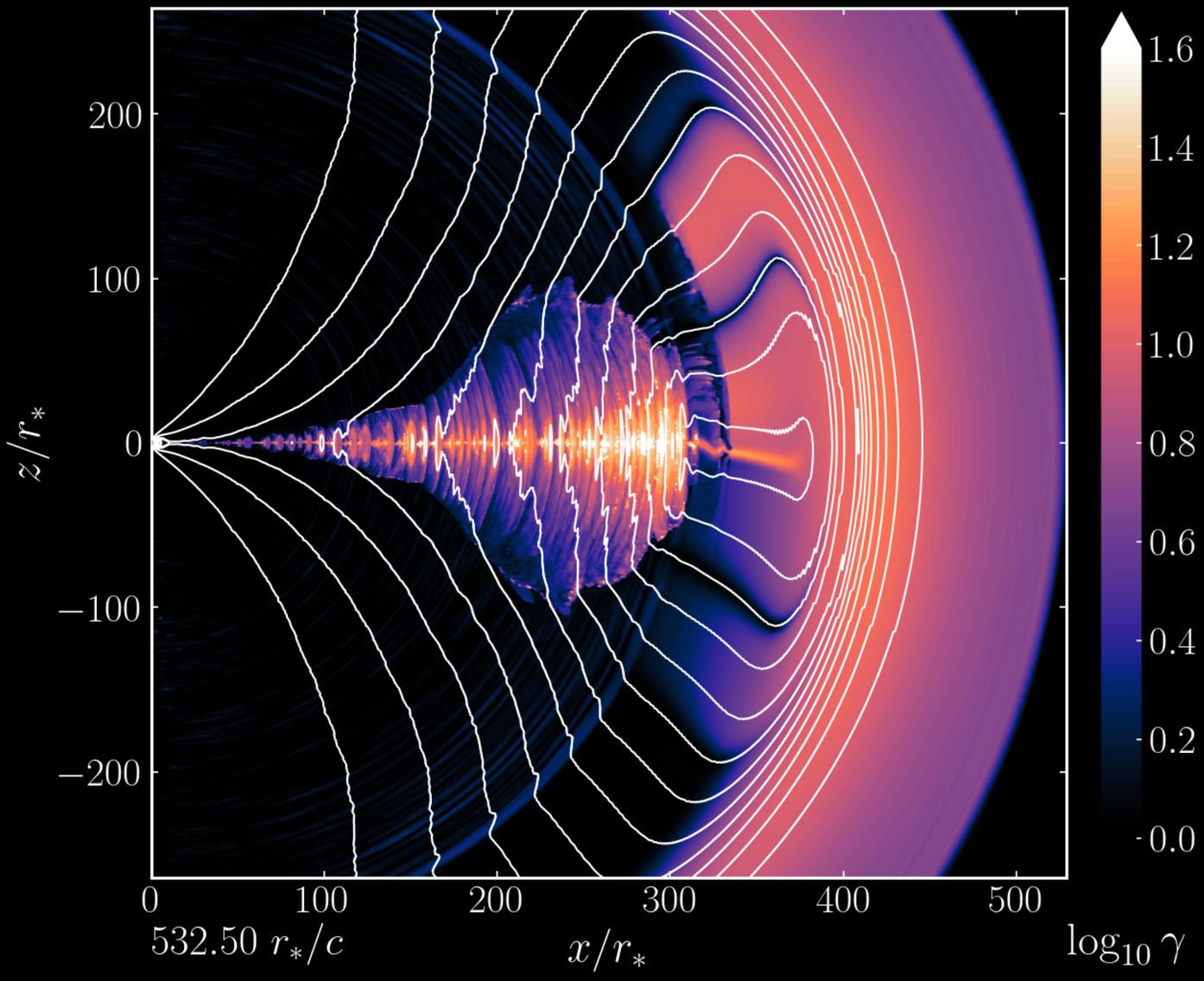
Giant flare X-ray, γ-rays ($\sim 10^{46}$ erg)

Remnant EM energy ($\sim 10^{45}$ erg)

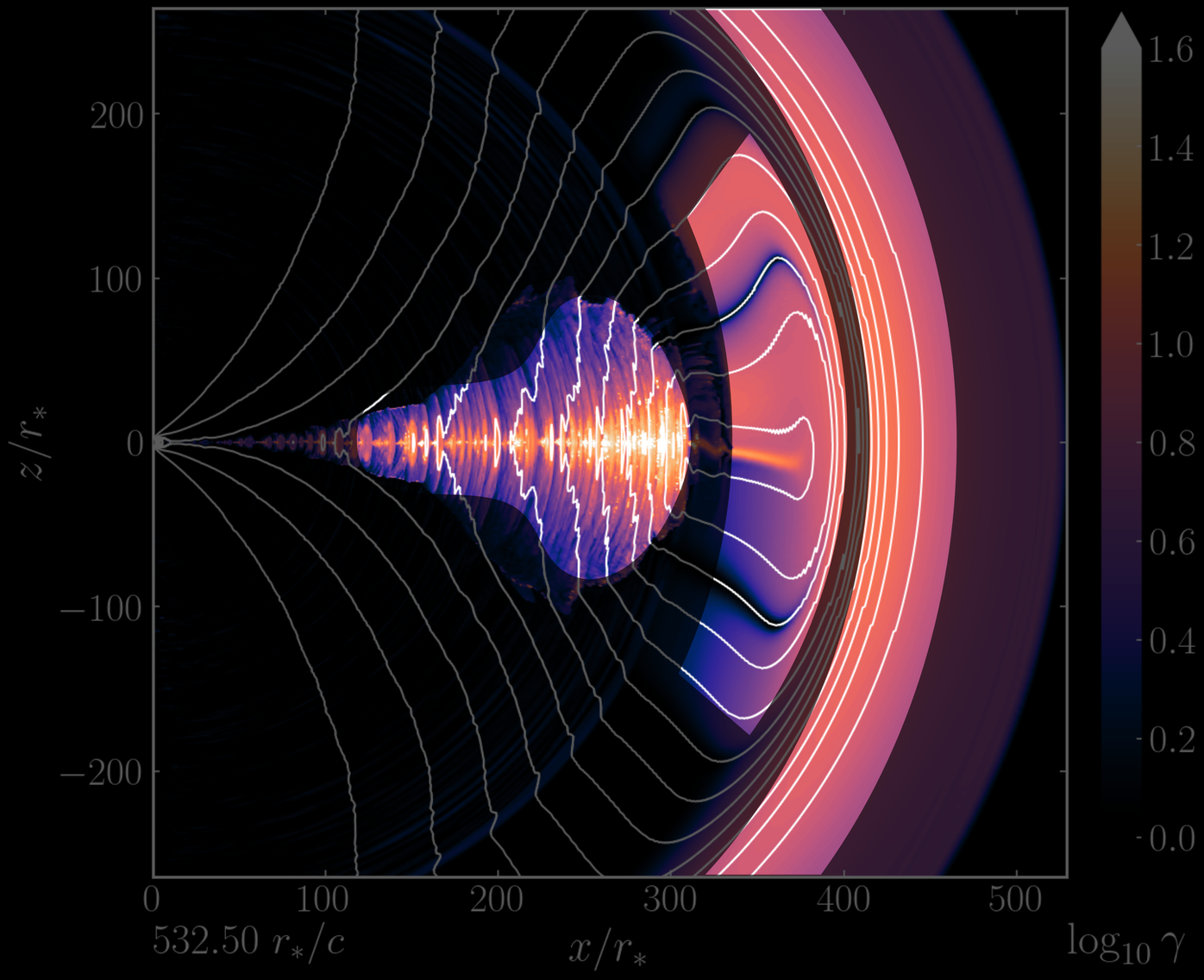
Fireball ($\sim 10^{45}$ erg)

*ejected flux is calculated at $30R_*$

Ejecta properties

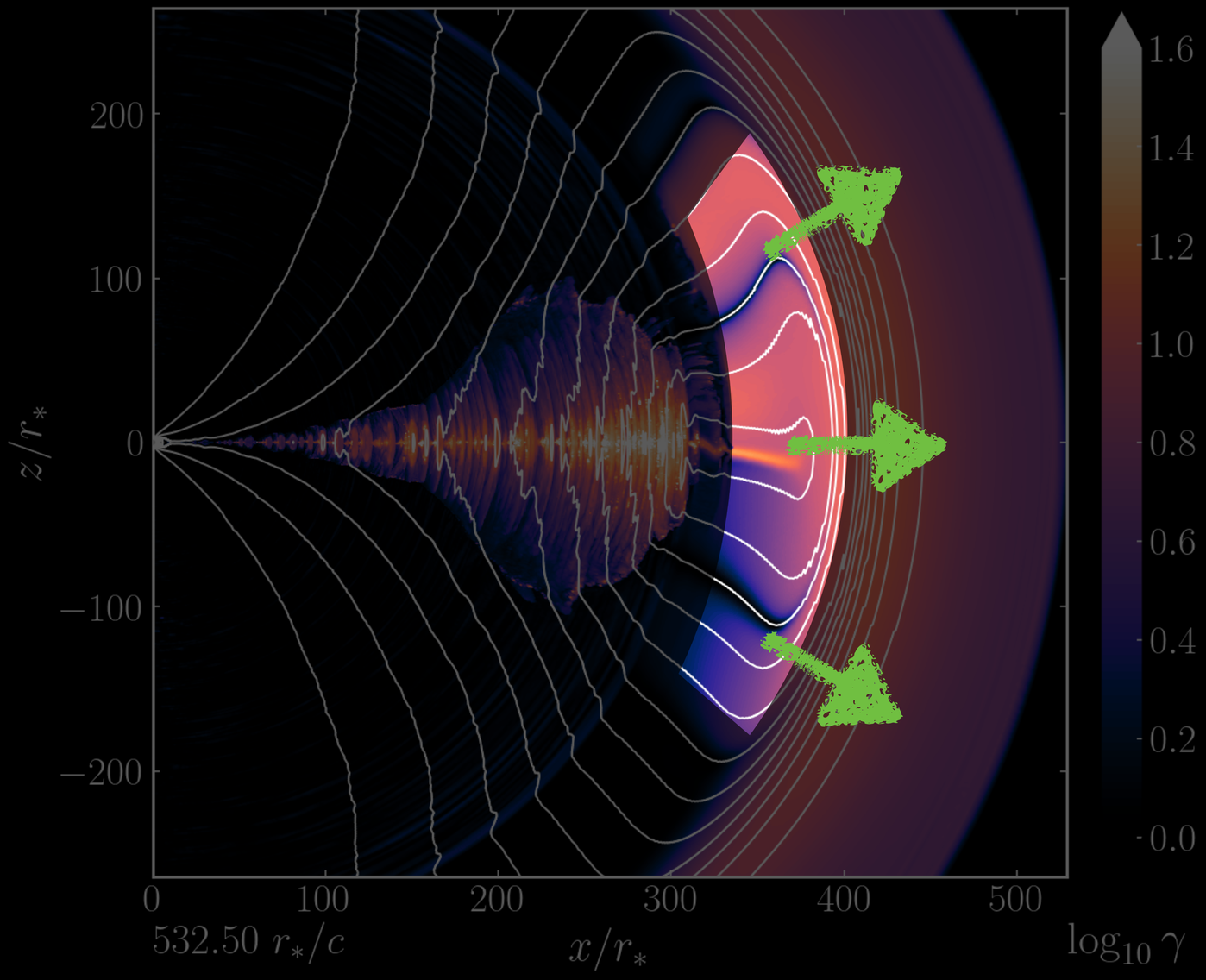


Ejecta properties



Ejecta properties

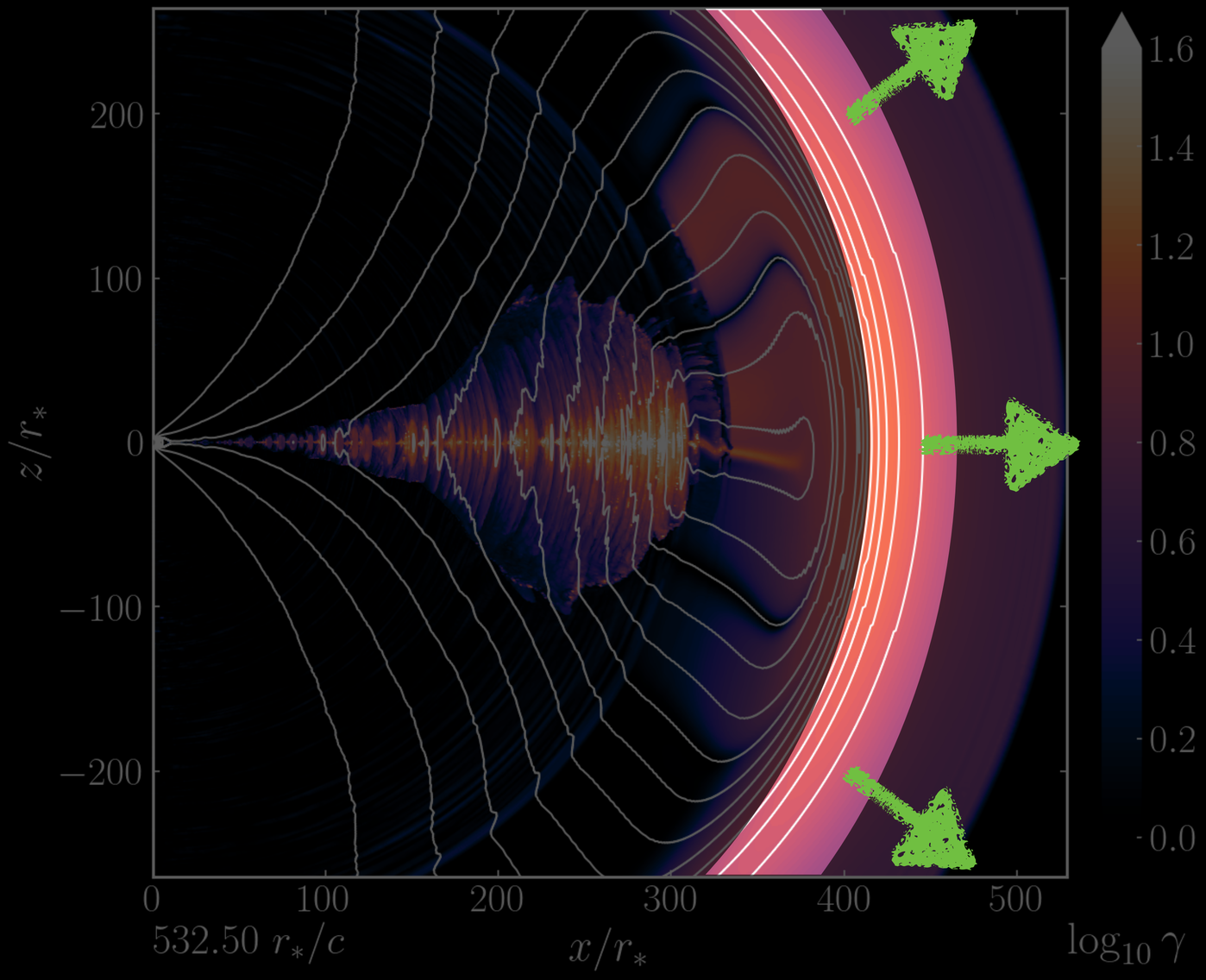
*assuming stellar B-field of $10^{15}\,\mathrm{G}$



Piston: Twisted fieldlines, cold plasma moving out at Lorentz factors $\gamma \gtrsim 14$; $E \sim 10^{46}\,\mathrm{erg}$

Ejecta properties

*assuming stellar B-field of $10^{15}\,\mathrm{G}$



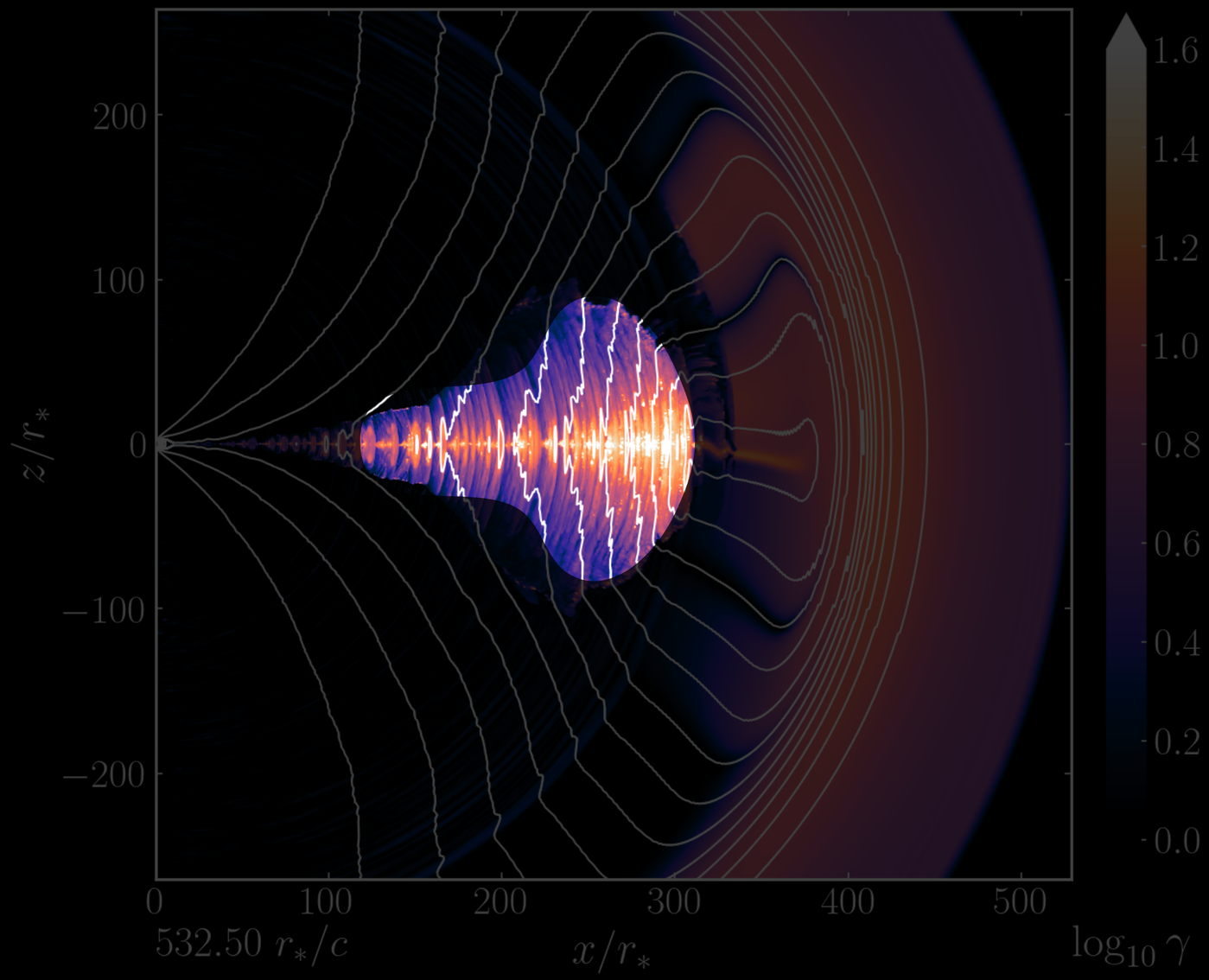
Piston: Twisted fieldlines, cold plasma moving out at Lorentz factors $\gamma \gtrsim 14$; $E \sim 10^{46}\,\mathrm{erg}$

EM pulse: compressed fields can power FRBs (~$10^{40}\,\mathrm{erg}$) *via shocks (Beloborodov+20), interactions with wind/current sheet (Lyubarsky+20)*

Pulse+Piston transitions from linear to slow acceleration phase when $\gamma \gtrsim \gamma_{\mathrm{Fast}} \sim 14$

Ejecta properties

*assuming stellar B-field of $10^{15}\,\mathrm{G}$



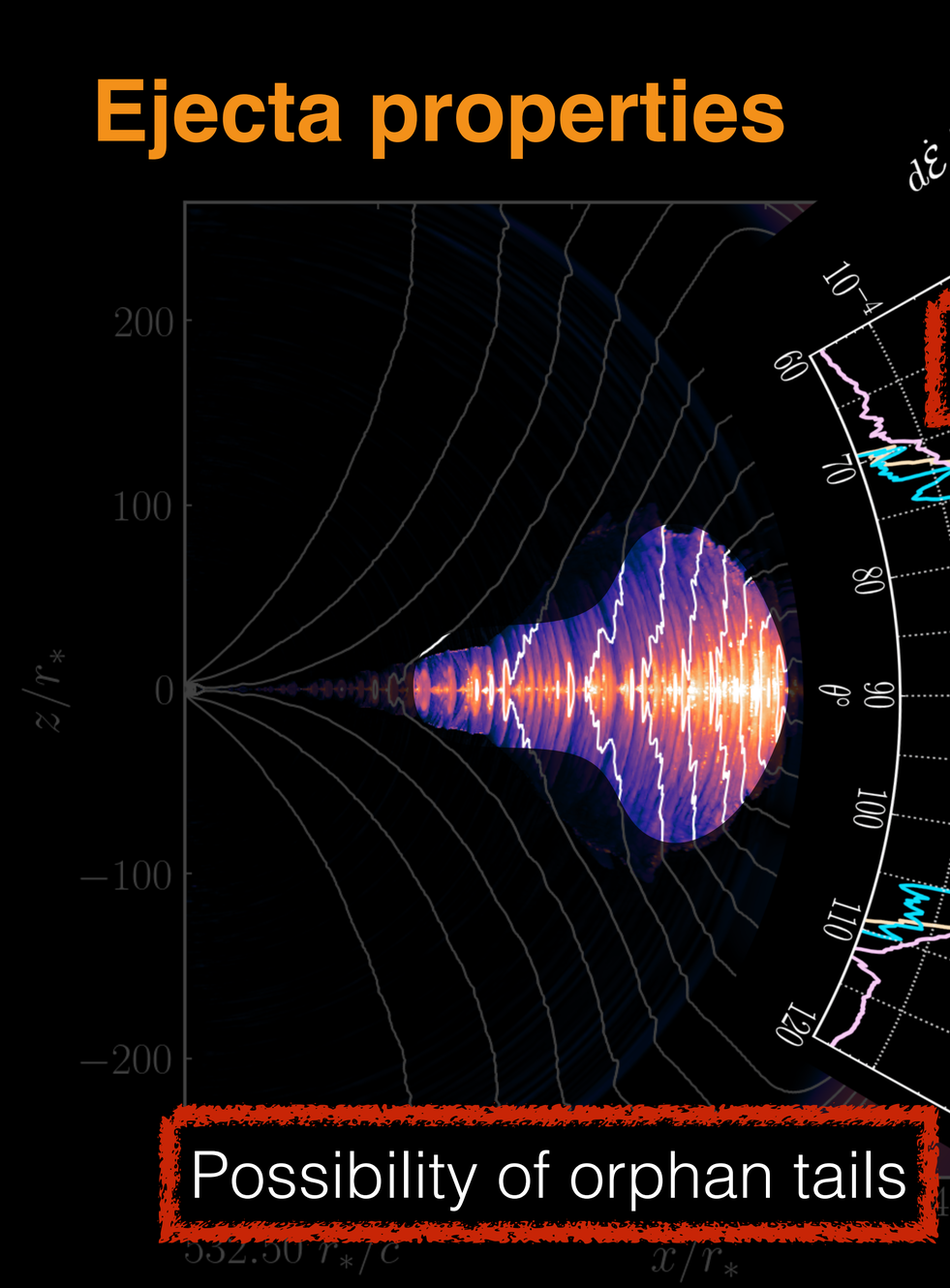
Piston: Twisted fieldlines, cold plasma moving out at Lorentz factors $\gamma \gtrsim 14$; $E \sim 10^{46}\,\mathrm{erg}$

EM pulse: compressed fields can power FRBs (~$10^{40}\,\mathrm{erg}$) *via shocks (Beloborodov+20), interactions with wind/current sheet (Lyubarsky+20)*

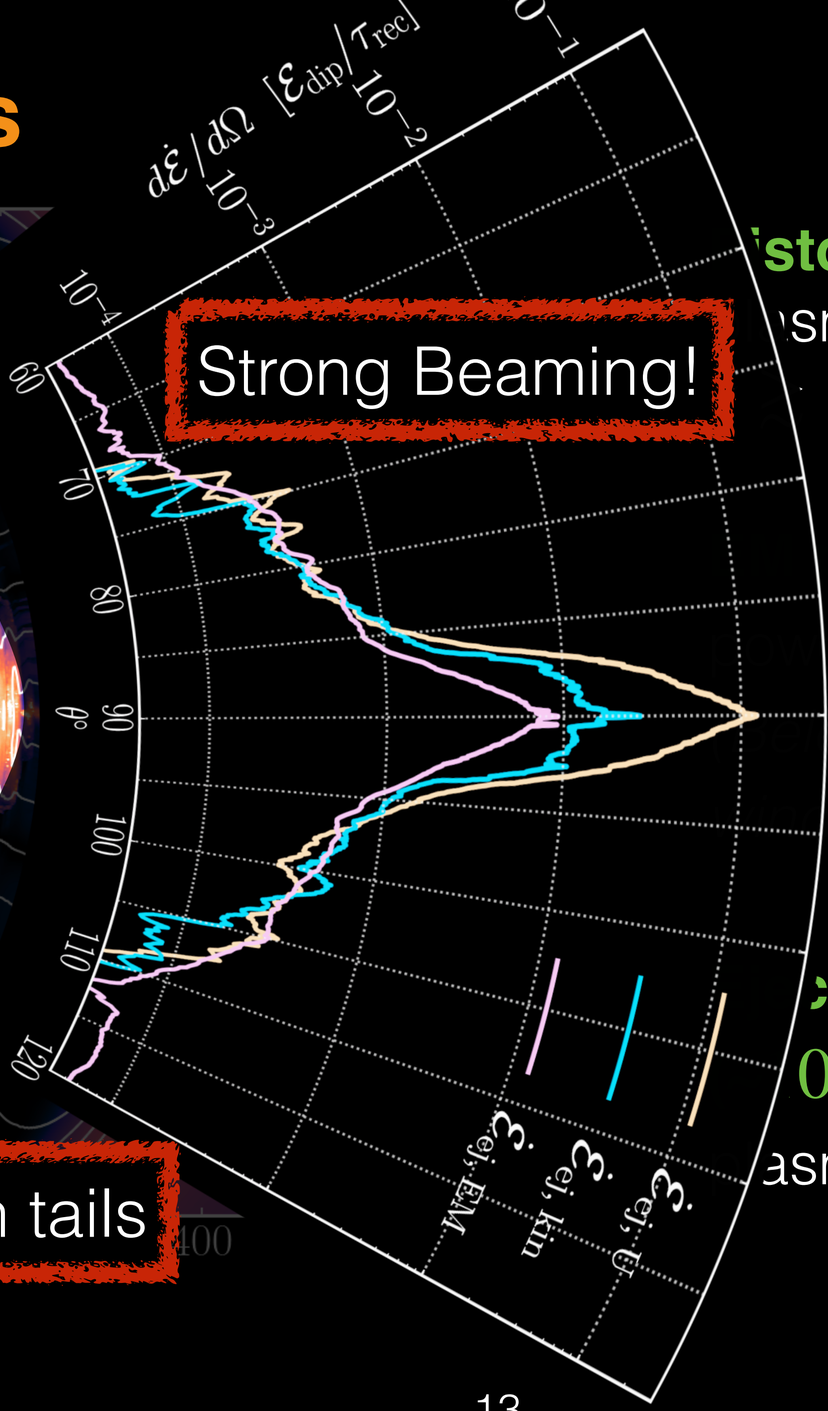
Ejecta tail or Escaping Fireball (~$10^{46}\,\mathrm{erg}$): filled with relativistic, hot plasma that sources X-ray, γ-rays

Ejecta properties

*assuming stellar B-field of $10^{15}\,\mathrm{G}$



Possibility of orphan tails



Strong Beaming!

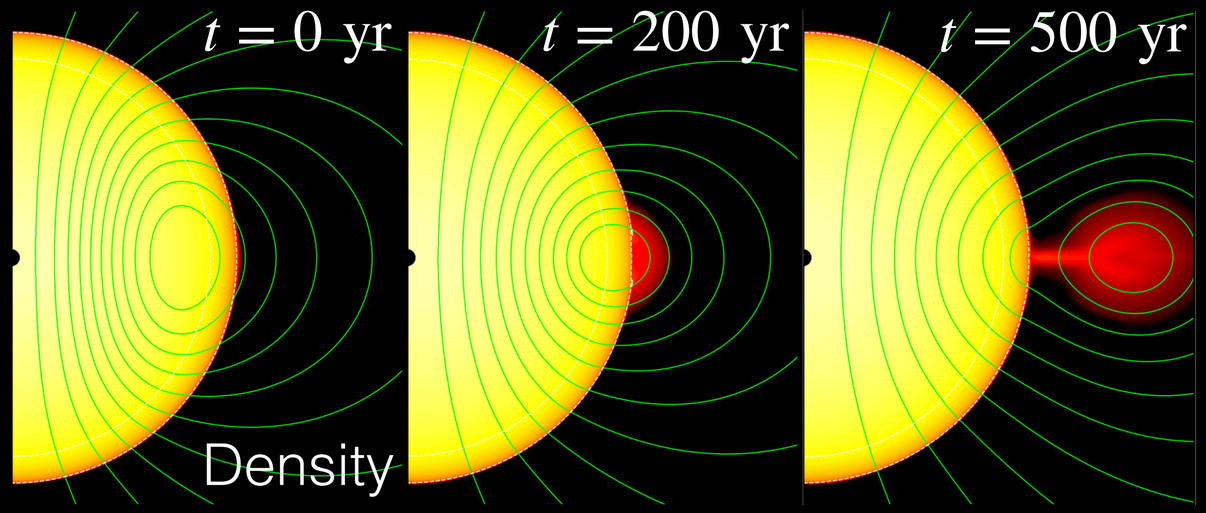
'ston: Twisted fieldlines, cold
sma moving out at Lorentz factors
14; $E \sim 10^{46}\,\mathrm{erg}$

ulse: compressed fields can
er FRBs (~$10^{40}\,\mathrm{erg}$) *via shocks*
borodov+20), interactions with
/current sheet (Lyubarsky+20)

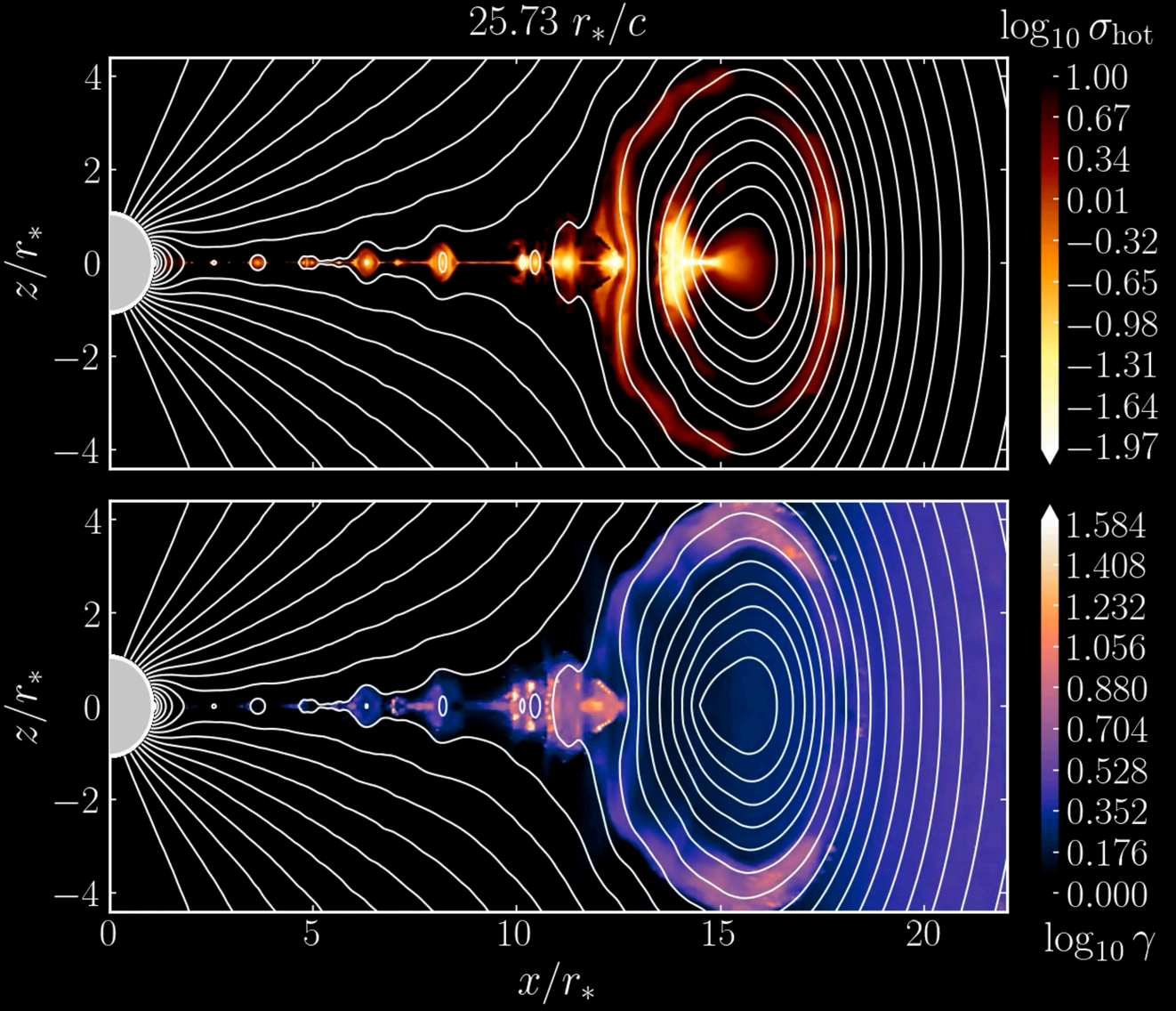
:ta tail or Escaping Fireball
$0^{46}\,\mathrm{erg}$**):** filled with relativistic, hot
asma that sources X-ray, γ-rays

Ejecta from baryon-rich bursts

KC, Bransgrove+, in prep



Bransgrove, Beloborodov & Levin 26



Magnetospheric model for crustal eruptions in hyperactive magnetars:

- relativistic outflow on ms timescales
- Trapped + Escaping fireballs
- Including gravity would slow down baryonic ejecta

*Preliminary models: no gravity!

Magnetar Giant Flares: Summary

With Grehan, Thompson, Ripperda

Results:

- Correct energy ranges (>10^{44} erg)
- Fireball distributions

Open questions:

- 3D effects: kinks, reconnection, beaming, NS interior dynamics

- Short flare durations (~few 10 ms):
  - Twists closer to the pole ($E \propto \sin^5\theta$)
  - Longterm twists: repeated eruptions
  - Radiative reconnection in the optically-thick regime with pair production

